Did Employer Skill Language Shift After the Generative AI Inflection Point?

Evidence from Job Postings Language in the U.S. (2019–2026)

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The “What”

Research Question

Main Question

Did employer skill language shift after the diffusion of generative AI in late 2022?

Sub-questions

- Did AI-related terminology increase?
- Did technical skill intensity change?
- Did emphasis on soft / coordination skills change?
- Did the overall topic structure of job descriptions shift?

Motivation

- Generative AI diffused rapidly after November 2022.
- Public debate focuses on employment loss and wages.
- Much less attention to how employer expectations evolve.
- Job postings provide real-time signals of labor demand.
- If technology restructures tasks, we should observe language shifts in job descriptions.

Theory

- Technological change restructures tasks, not just employment (Autor et al., 2003).
- Automation substitutes routine work but increases demand for complementary skills (Acemoglu & Restrepo, 2020).
- Technology-intensive production raises demand for cognitive and coordination skills (Deming, 2017).
- If generative AI automates routine cognitive tasks, employer skill language should shift after late 2022:

  ↑ AI task language

  ↑ Complementary skills

  ↓ Routine cognitive tasks

The "How"

Identification Strategy

- Generative AI diffusion represents a common technological shock.
- Occupations differ in their pre-existing exposure to AI capabilities.
- Late 2022 marks a visible acceleration in public adoption and organizational awareness of generative AI tools such as release of ChatGPT.

Exposure Measure: **AI Occupational Exposure Index (AIOE)**

- Developed by Felten, Raj & Seamans (2021)
- Using O*NET data links AI applications to occupational abilities
- Occupation-level (SOC codes)
- Time-invariant
- AIOE index is a standardized measure (essentially a Z-score).
- Continuous (-2.67 to 1.53), here mean is 0.
- Independent of post-2022 outcomes

Ranking of Occupational Categories by AIOE Score (average)

Category Name	AIOE
Legal Occupations	1.31
Business and Financial Operations Occupations	1.17
Computer and Mathematical Occupations	1.11
Educational Instruction and Library Occupations	0.99
Management Occupations	0.96
Community and Social Service Occupations	0.88
Life, Physical, and Social Science Occupations	0.83
Architecture and Engineering Occupations	0.81
Office and Administrative Support Occupations	0.65
Sales and Related Occupations	0.64
Arts, Design, Entertainment, Sports, and Media Occupations	0.27

Category Name	AIOE
Healthcare Practitioners and Technical Occupations	0.19
Personal Care and Service Occupations	- 0.29
Protective Service Occupations	- 0.36
Healthcare Support Occupations	- 0.51
Transportation and Material Moving Occupations	- 0.74
Food Preparation and Serving Related Occupations	- 0.77
Production Occupations	- 0.83
Installation, Maintenance, and Repair Occupations	- 0.88
Farming, Fishing, and Forestry Occupations	- 1.06
Building and Grounds Cleaning and Maintenance Occupations	- 1.15
Construction and Extraction Occupations	- 1.34

Empirical Design

- Common shock: Generative AI diffusion
- Treatment intensity: AIOE (continuous, occupation-level)
- Unit of analysis: Occupation × Month
- Method: Continuous Difference-in-Differences
- Shock timing : Nov 2022
- Pre: 2019 –2022
- Post: 2023 – 2026

Variables

- **Outcome variables:**

Variables used	Operationalization	Keywords used (sample)
1. AI keyword density	$\frac{Number\ of\ AI\ Keywords\ mentions\ in\ the\ job\ title\ \&\ description}{Number\ of\ words\ in\ job\ description}$	artificial intelligence, AI, machine learning, deep learning, NLP, LLM, generative AI, ChatGPT, automation, algorithm, predictive, modeling, and model.
2. Technical skill density	$\frac{Number\ of\ Technical\ skill\ Keywords\ mentions\ in\ the\ job\ title\ \&\ description}{Number\ of\ words\ in\ job\ description}$	data, analytics, analysis, statistics, Python, SQL, database, cloud, software, programming, coding, cybersecurity, information systems, machine learning, algorithm, technical, technology, dashboard, visualization, and modeling.
3. Soft skill density	$\frac{Number\ of\ Soft\ skill\ Keywords\ mentions\ in\ the\ job\ title\ \&\ description}{Number\ of\ words\ in\ job\ description}$	Communication, collaboration, team, teamwork, leadership, manage, management, coordinate, problem solving, critical thinking, adaptability, interpersonal, customer service, support, relationship, presentation, and writing.
The dictionaries are not meant to be mutually exclusive. They capture different dimensions of job-posting language. Some terms may reasonably belong to both AI-related and technical-skill language.		

- **Controls: No additional job-level controls; model specification relies on year structure and AIOE-based exposure due to limited uniform covariates.**

Baseline DiD model

$$\boldsymbol{Y_{it} = \alpha + \beta\ (\ Post_{t+1} \times AIOE_i\ ) + \gamma_i + \delta_t + \epsilon_{it}}$$

Where:

$\boldsymbol{Y_{it}}$ = skill intensity or topic prevalence

$\boldsymbol{Post_{t+1}}$ = 1, shock + 1 year (2023 - 2026)

$\boldsymbol{AIOE_i}$ = occupation-level AI exposure

$\boldsymbol{\gamma_i}$ = occupation fixed effects

$\boldsymbol{\delta_t}$ = year fixed effects

β measures whether occupations with higher AI exposure experienced larger post-2022 shifts in employer skill language.

Exposure is continuous to preserve cross-occupational variations.

Event study (Parallel Trends & Dynamics)

$$Y_{it} = \alpha + \sum_{\{k \neq -1\}} \beta_k \left(D_t^k \times AIOE_i \right) + \gamma_i + \delta_t + \varepsilon_{it}$$

Rationale:

- Test for differential pre-trends
- Examine timing of language shifts
- Observe whether restructuring is immediate or gradual
- This allows us to assess whether divergence emerges only after late 2022.
- Omitted period: 2022.

The "Mining"

Job Postings Data Pipeline & Sample Construction

RAW COLLECTION

① Raw Job Postings Collected		
Year	Platform	Sample
2019	Indeed	~35,000
2020	Indeed	~35,000
2021	Indeed	~30,000
2022	Indeed, Google, OSF, Upwork	~7,500
2023	LinkedIn	~16,000
2024	Indeed, LinkedIn, Upwork	~150,000
2025	Indeed, LinkedIn	~35,000
2026	Greenhouse, Jobicy, Lever, LinkedIn, Muse, USA Jobs	~20,000
Total		~328,500

AIOE MATCHING

② Match with AIOE Categories

60 – 70%

match rate

Unique AIOE categories matched

FINAL SAMPLE

③ Analysis Sample			
Year	Platform	Sample	AIOE Cats
2019	Indeed	26,094	17
2020	Indeed	29,967	16
2021	Indeed	29,939	17
2022	Indeed	Excluded	
2023	LinkedIn	13,443	22
2024	Indeed	14,362	22
2025	Indeed	36,107	22
2026	Indeed	Excluded	
Total Sample Size 149,912			

Note: 2022 & 2026 data excluded from final sample due to low volume after AIOE matching | AIOE = Automation and Occupational Employment categories

Knowledge Mining Techniques

1. TEXT PROCESSING

Standardize raw job posting text into analysis-ready tokens

- Cleaning
- Tokenization
- Lemmatization

2. FEATURE CONSTRUCTION

Build rich feature vectors capturing skill demand signals

- AI-specific keyword dictionary
- Technical skill dictionary
- Soft skill dictionary
- TF-IDF weighting

3. ADVANCED ANALYSIS

Uncover latent skill clusters and measure AI-era shifts

- Topic Modeling (LDA)
- Topic prevalence comparison (Pre vs. Post)

AI Assistance

What we used	
Claude Sonnet 4.6	Coding
Gemini 3.1	Brainstorming, Literature summaries, Presentation, Language check and polishing etc.
ChatGPT-5.5	

How we used them

1. Brainstorming & Ideation
Polish research questions & hypotheses
Explore novel angles on AI labor market impact
Summarizing prior research work

2. Data Sources & Links
Surface relevant datasets (BLS, O*NET, AIOE)
Recommend job posting platforms to scrape
Point to open-access repositories & APIs

AI tools used as research accelerators — all analysis, interpretation & conclusions reflect the authors' own scholarly judgment

AI Assistance

How we used them

3. Literature & Prior Research

- Summarize key papers on automation & skills
- Compare methodologies across studies
- Flag seminal works & citation chains

4. Code Writing & Debugging

- Write Python/R for data cleaning & NLP pipeline
- Build LDA topic models & TF-IDF matrices
- Debug errors & optimize performance

5. Statistical Interpretation

- Examine regression & model outputs
- Suggest robustness checks & sensitivity tests
- Validate significance & effect sizes

6. Writing & Editing

- Checking initial drafts
- Suggest improvements
- Explore transitions & argument structures

AI tools used as research accelerators — all analysis, interpretation & conclusions reflect our own scholarly judgment

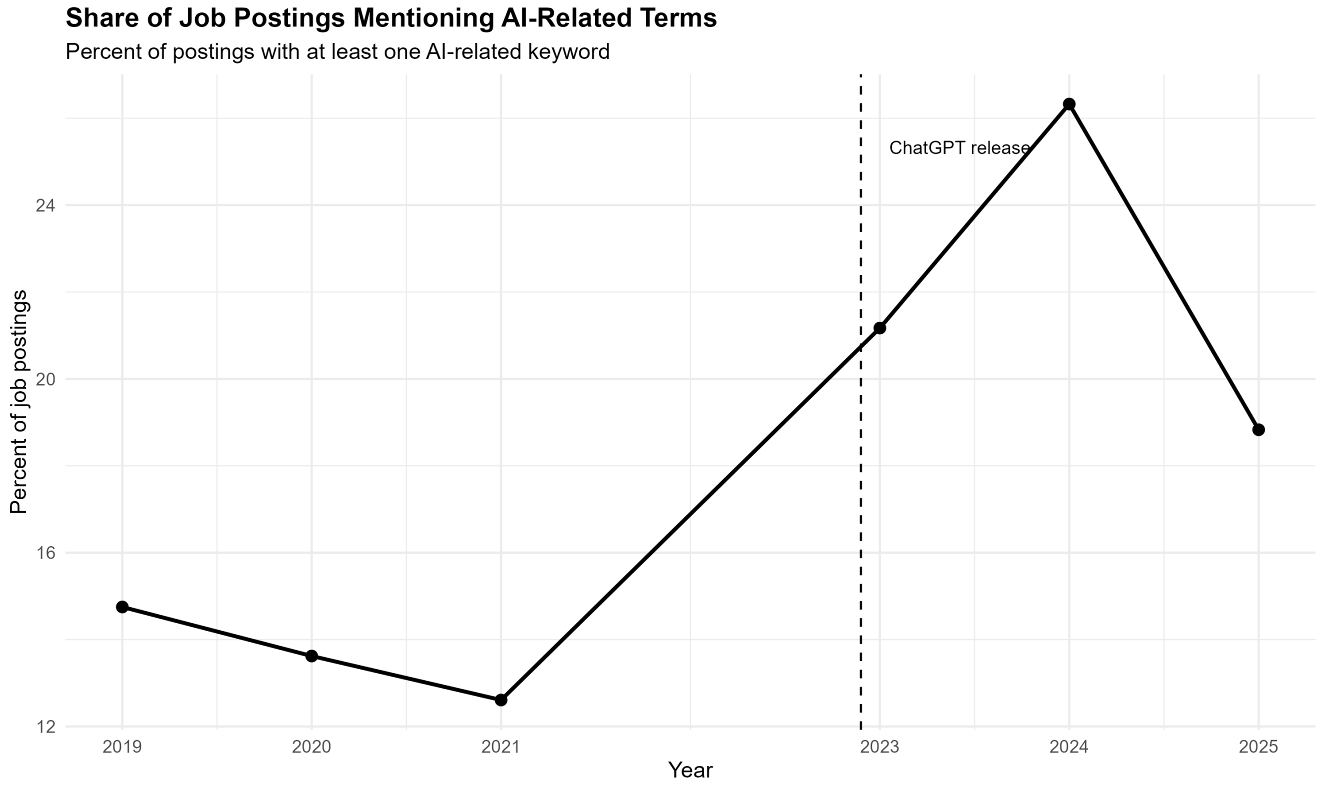
Text Mining Pipeline and Outcome Variables

Task	***Method***
A. Combine text	Merged job title + description using mutate() / paste()
B. Clean text	Lowercased, removed symbols, standardized spacing using stringr
C. Tokenize text	Split postings into words using tidytext::unnest_tokens()
D. Apply dictionaries	Matched AI, technical-skill, and soft-skill keyword lists
E. Create outcomes	Counted keyword hits and calculated keyword density
F. Export file	Saved modeling-ready job-level outcomes using write_csv()

Outcome variables: AI keyword density, technical-skill density, and soft-skill density.

AI Language in Job Postings

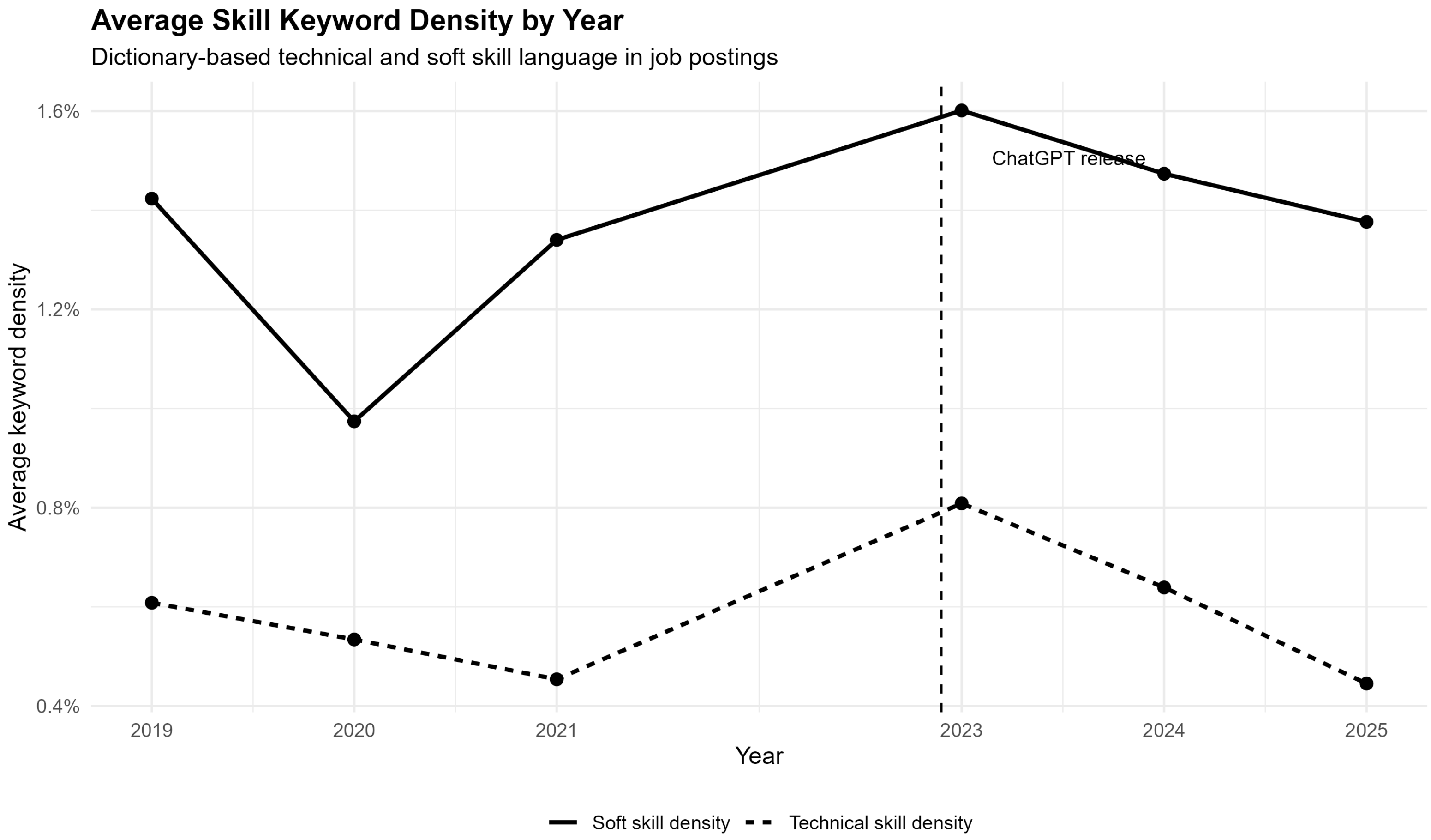
- Measures the share of job postings with at least one AI-related keyword
- AI mentions declined slightly from 2019 to 2021
- Mentions increased clearly in 2023 and peaked in 2024
- 2025 declined from 2024, but stayed above the 2021 level



AI-related language became more visible in job postings after the ChatGPT period, especially in 2023 - 2024.

Skill Language in Job Postings

- Compares average keyword density for soft skills and technical skills
- Soft-skill language is consistently more common than technical-skill language
- Both types of skill language rise after 2021 and peak around 2023
- Both decline after 2023, especially technical-skill density

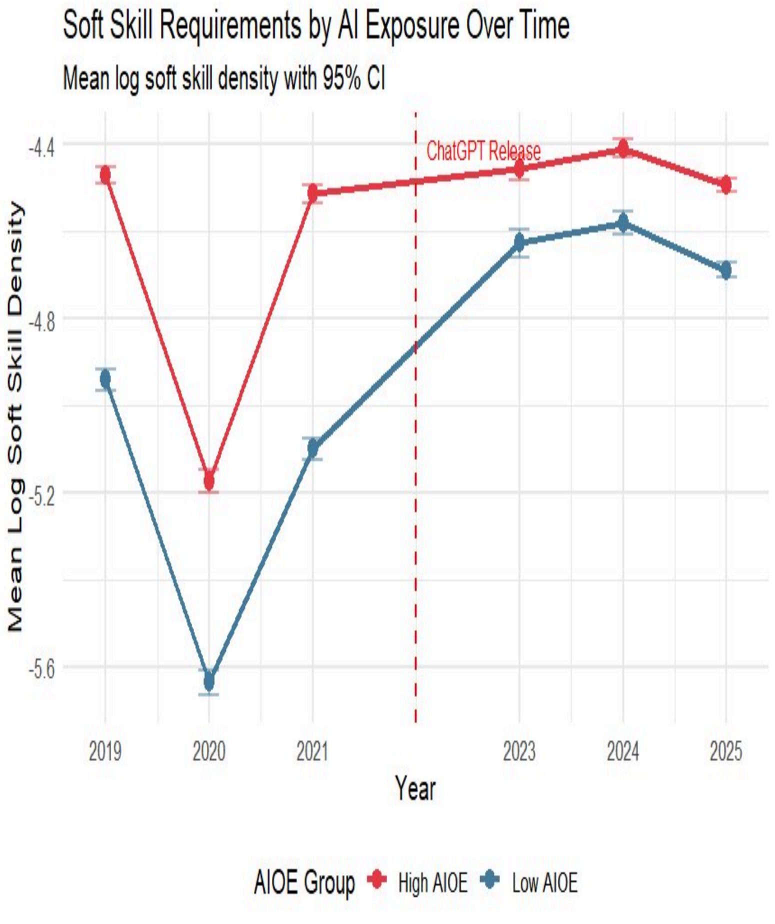


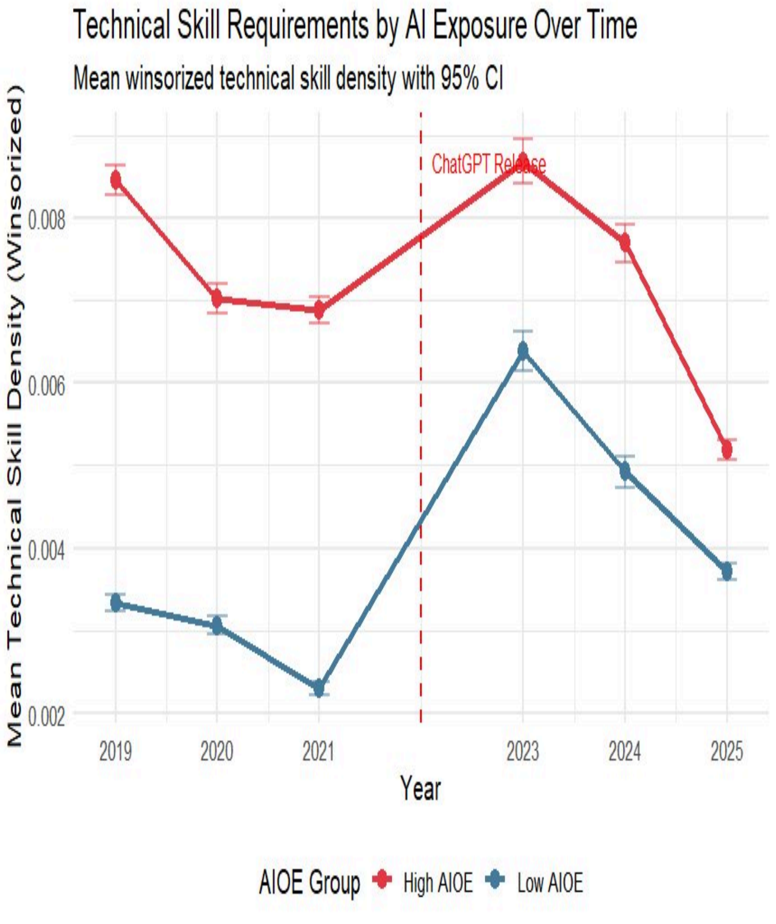
Job postings emphasize soft skills more than technical skills, but both types of language shift over time.

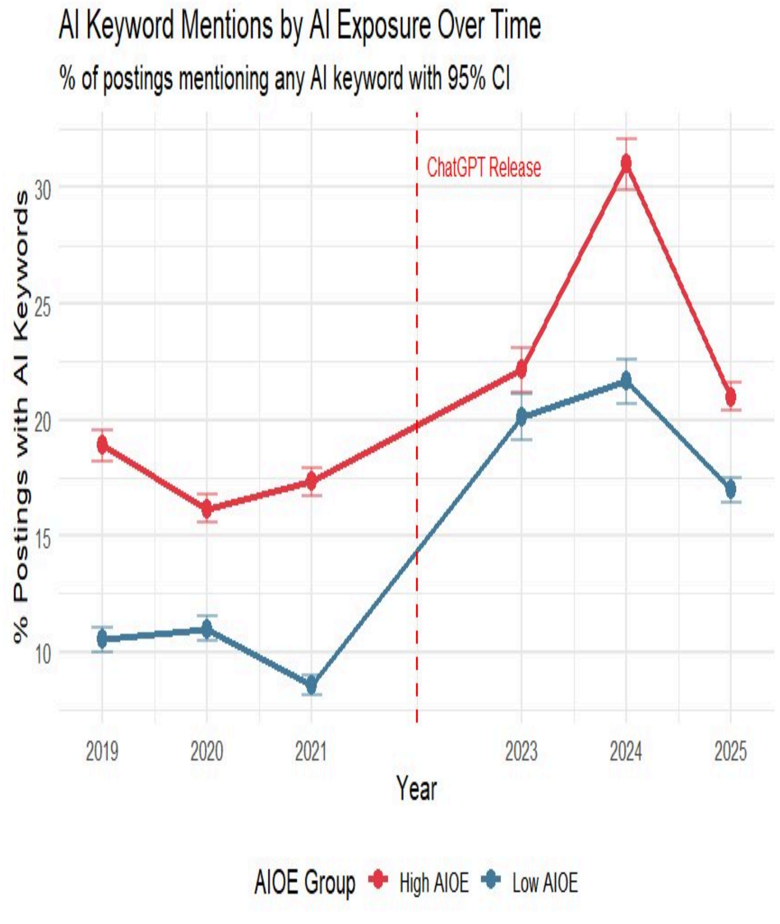
The Evidence

Descriptive Trends: Skill Requirements by AI Exposure

Pre vs. Post ChatGPT Release (Nov 2022) · High AIOE vs. Low AIOE Jobs · N = 149,912 job postings



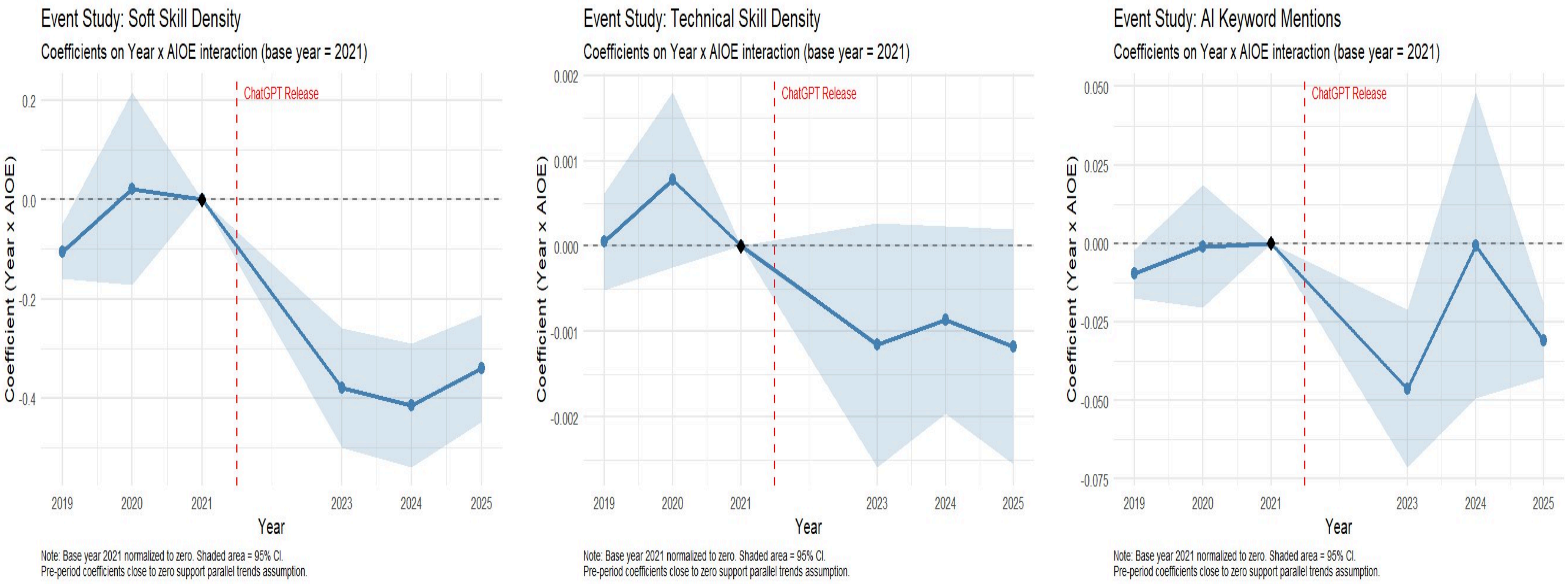




Key Finding: After ChatGPT, Low AIOE jobs converge toward High AIOE jobs in soft skills & AI keyword usage. High AIOE jobs show declining technical skill density post-2022.

Parallel Trends Assumption: Event Study

Year × AIOE interaction coefficients · Base year = 2021 · Pre-period coefficients near zero validate DiD assumption



Pre-Period (2019–2021): Coefficients close to zero with CI crossing zero → groups moved in parallel before ChatGPT

Post-Period (2023–2025): Coefficients diverge below zero → significant differential effect after ChatGPT release

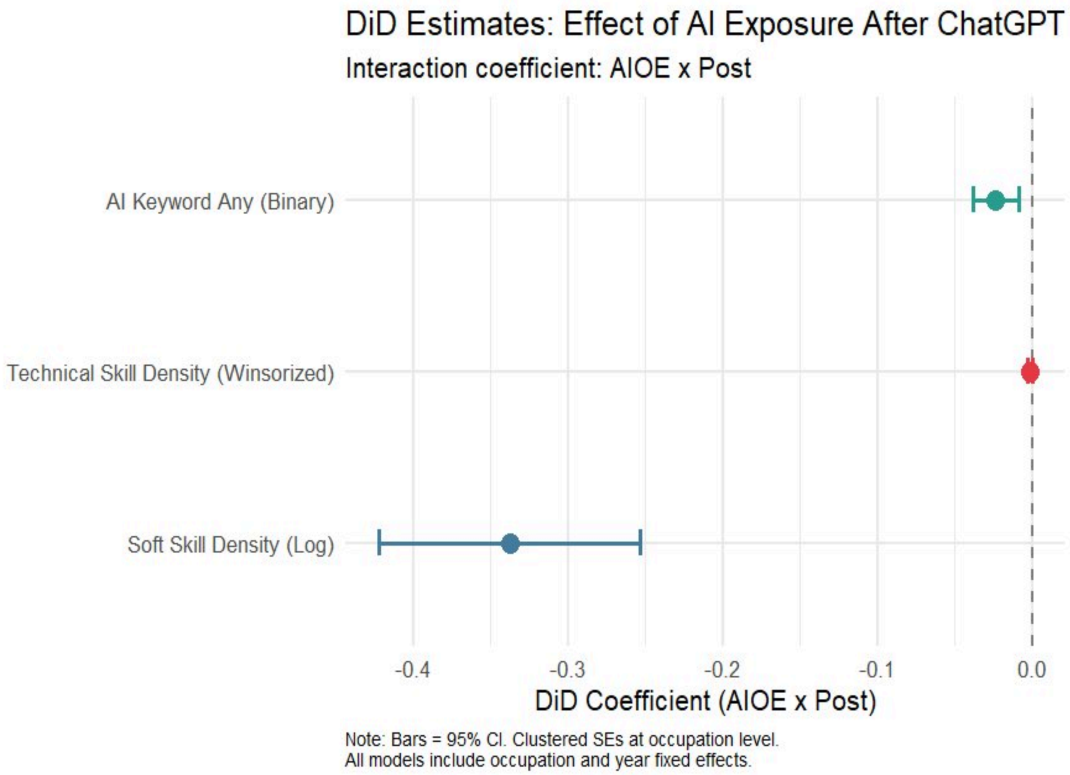
Difference-in-Differences Results

AIOE × Post interaction · Occupation & Year Fixed Effects · Clustered SEs at occupation level · N = 149,912

Regression Table

Outcome Variable	Soft Skill (Log)	Technical Skill (W)	AI Keyword (Binary)
AIOE × Post (DiD)	**-0.3378***** (0.0407)	**-0.0014*** (0.0006)	**-0.0236**** (0.0071)
Fixed Effects			
Occupation FE			
Year FE			
Observations	149,912	149,912	149,912
R^2	0.096	0.166	0.045
Clustered SEs	Occupation	Occupation	Occupation

**** p<0.001   ** p<0.01   * p<0.05 | SE in parentheses*



Soft Skill Density

β = −0.338***
Higher AI-exposed jobs show relatively lower soft skill density

Technical Skill Density

β = −0.0014* (−26.6%)
Higher AI-exposed jobs show relatively lower technical skill density

AI Keyword Mentions

β = −0.0236 (−6.3pp)**
Higher AI-exposed jobs are relatively less likely to mention AI keywords

Conclusions & Implications

How did ChatGPT's release change language skill requirements across AI-exposed and non-AI-exposed jobs?

What We Found

Higher AI-exposed jobs show relatively lower soft skill density after ChatGPT

Higher AI-exposed jobs show relatively lower technical skill density after ChatGPT

Higher AI-exposed jobs are relatively less likely to mention AI keywords after ChatGPT

What This Means

AI Substitution: ChatGPT appears to substitute for skills rather than complement them in high AI-exposed jobs

Skill Convergence: Lower AI-exposed jobs are catching up — narrowing the skill gap across occupations

Normalization: Employers in high AI-exposed jobs may assume AI handles certain tasks — reducing explicit skill demands

Limitations

- Only 22 occupation clusters (AIOE categorical was used) — standard errors may be conservative
- Year 2022 is missing from the data — the exact year of ChatGPT's release — preventing direct observation of the immediate shock period

Future Research

- Monthly data to capture precise timing of skill adjustments
- Occupation-level heterogeneity within AIOE groups
- Vocabulary analysis to identify specific keywords driving changes

Thank You

We welcome your questions, comments, and suggestions.